UNIT IV

ASSIGNMENT

Class #1:

Orthogonal Bases- Orthogonal Matrices, Properties,

1) Let $v_1 = \begin{pmatrix} -2 \\ 2 \\ 1 \\ 4 \end{pmatrix}, v_2 = \begin{pmatrix} 4 \\ 1 \\ -2 \\ 2 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 4 \\ 2 \\ -2 \end{pmatrix}$ and $B = \{v_1, v_2, v_3\}$ is an orthogonal set W be the subspace spanned by $\{v_1, v_2, v_3\}$.

a) Find the orthogonal projection of $\begin{pmatrix} -13 \\ 8 \\ 9 \\ 1 \end{pmatrix}$ into w.

ANS: $p = \begin{pmatrix} -11 \\ 16 \\ 13 \\ 2 \end{pmatrix}$

b) Find an orthonormal basis for w.

ANS: $\left\{ \begin{pmatrix} \frac{-4}{5} \\ \frac{1}{5} \\ \frac{-2}{5} \\ \frac{7}{5} \end{pmatrix} \begin{pmatrix} \frac{1}{5} \\ \frac{4}{5} \\ \frac{2}{5} \\ \frac{-2}{5} \end{pmatrix} \begin{pmatrix} \frac{-2}{5} \\ \frac{2}{5} \\ \frac{1}{5} \\ \frac{4}{5} \end{pmatrix} \right\}$

2) Let w be the subspace spanned by $\left\{ \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \begin{pmatrix} 0 \\ -3 \\ 0 \\ -3 \end{pmatrix} \begin{pmatrix} 0 \\ 5 \\ 5 \\ 7 \end{pmatrix} \right\}$ and this basis not an orthogonal.

a) Find the vector in W closed to $\begin{pmatrix} -1 \\ 7 \\ -5 \\ -4 \end{pmatrix}$

ANS: $\begin{pmatrix} 0 \\ 7 \\ -5 \\ -4 \end{pmatrix}$

b) Find an orthonormal basis for w.

ANS: $\left\{ \begin{pmatrix} 0 \\ -2 \\ -2 \\ -1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 2 \\ -2 \end{pmatrix} \begin{pmatrix} 0 \\ -2 \\ 1 \\ 2 \end{pmatrix} \right\}$

3) Show that $\{v_1, v_2, v_3\}$ is an orthogonal set where,

$v_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} v_{2=} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} v_2 = \begin{pmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{pmatrix}$

4) The set $\{v_1, v_2, v_3\}$ is an orthogonal basis in R^3. Express the vector $y = \begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix}$ as a linear combination of the vector in S, where $v_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} v_{2=} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} v_2 = \begin{pmatrix} -\frac{1}{2} \\ -2 \\ \frac{7}{2} \end{pmatrix}$

ANS: $y = v_1 - 2v_3 - 3v_3$

5) Show that $\{v_1, v_2, v_3\}$ is an orthonormal basis in R^3, where

$v_1 = \begin{pmatrix} \frac{3}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \end{pmatrix} v_{2=} \begin{pmatrix} \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix} v_2 = \begin{pmatrix} \frac{-1}{\sqrt{66}} \\ \frac{-1}{\sqrt{66}} \\ \frac{7}{\sqrt{66}} \end{pmatrix}$

6) Let $y_1 = \begin{pmatrix} \frac{-1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$, $y_2 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$, show that $\{y_1, y_2\}$ is an orthonormal basis for R^2

7) Show that $\{v_1, v_2\}$ *or* $\{v_1, v_2, v_3\}$ is an orthogonal basis for R^2 or R^3. Then express x as the linear combination of the $v's$

(i) $v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} v_{2=} \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} v_2 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, x = \begin{pmatrix} 8 \\ -4 \\ 3 \end{pmatrix}$

(ii) $v_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix} v_{2=} \begin{bmatrix} 6 \\ 4 \end{bmatrix} x = \begin{bmatrix} 9 \\ -7 \end{bmatrix}$

8) Let u and v be orthogonal vectors. If u + v and u − v are orthogonal, show that |u| = |v|.